

EXPERIMENT – 9

ULTRASONIC SENSOR USING ARDUINO IN PROTEUS

```
ultrasonic.ino
1 // Program Sketch
2 int trig = 10;
3 int echo = 9;
4 long duration;
5 int cm;
6 void setup() {
7 // put your setup code here, to run once:
8 pinMode(trig, OUTPUT);
9 pinMode(echo, INPUT);
10 Serial.begin(9600);
11 }
12 void loop() { // Sketch to calculate ultrasonic distance
13 digitalWrite(trig, LOW);
14 delayMicroseconds(10);
15 digitalWrite(trig, HIGH);
16 delayMicroseconds(10);
17 digitalWrite(trig, LOW);
18 delayMicroseconds(10);
19 duration = pulseIn(echo, HIGH);
20 cm = (duration/2) * 0.034;
21 // Print Value on Serial Monitor
22 Serial.print(" Distance = ");
23 Serial.print(cm);
24 Serial.println(" cm");
}
```

Output

```
jino15\packages\arduino\tools\avr-gcc\7.3.0-atmel3.6.1-arduino7\bin/avr-objcopy -O ihex -j .eeprom --set-section-flags=.eeprom=alloc,load --no-change-wa
jino15\packages\arduino\tools\avr-gcc\7.3.0-atmel3.6.1-arduino7\bin/avr-objcopy -O ihex -R .eeprom "C:\Users\Jerolin Mary\AppData\Local\arduino\sk
jino15\packages\arduino\tools\avr-gcc\7.3.0-atmel3.6.1-arduino7\bin/avr-size" -A "C:\Users\Jerolin Mary\AppData\Local\arduino\sketches\20597569F0B
: space. Maximum is 32256 bytes.
ic memory, leaving 1842 bytes for local variables. Maximum is 2048 bytes.
```

